

Subin Biju

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Professional Summary

AI/ML Engineer with a B.Sc. in Computer Science and Post Graduate Diploma in Artificial Intelligence and Machine Learning. Hands-on experience building end-to-end AI applications using Python, machine learning, deep learning, computer vision, LLM/VLM integration, REST APIs, and data pipelines. Experienced with RAG concepts, AI agents, prompt engineering, vector search, FastAPI, PostgreSQL, Redis/RQ, Docker, Git, Streamlit, and local LLM/VLM deployment.

Technical Skills

Programming: Python, Data Cleaning, Data Preprocessing

AI/ML: Machine Learning, Deep Learning, Regression, Classification, Clustering, Random Forest, CNN, RNN, Transformers, Model Deployment

Generative AI: LLMs, LLM APIs, Prompt Engineering, Qwen2-VL, Vision-Language Models, LLM Integration, AI Workflows

AI Engineering: RAG, Embeddings, Semantic Search, Vector Search, AI Agents, Multi-Agent Concepts

Frameworks: Scikit-learn, TensorFlow, PyTorch, Keras, OpenCV

Backend & APIs: FastAPI, Flask, REST APIs, API Integration

Data & Infrastructure: PostgreSQL, MySQL, Redis, RQ, Docker, Git, GitHub, Jupyter, Google Colab, LM Studio

Projects

Smart Food Guardian — AI-Powered Food Health Scanner

Aug 2025 – Sep 2025

- Built a privacy-first AI pipeline using Python, OpenCV, Qwen2-VL, and Random Forest to extract food-label information and generate personalized health insights, achieving 84% accuracy across 181 products.
- Integrated a local Vision-Language Model through LM Studio and Streamlit for fully local AI inference, reducing dependency on cloud-based processing.
- Implemented decision logic for allergen alerts and goal-based health insights from extracted product information.

AI-Powered Application Generator

Aug 2024 – Dec 2024

- Developed an LLM-powered system that generates application code from natural-language prompts, demonstrating LLM integration and AI-driven automation.
- Built REST APIs using FastAPI with asynchronous AI processing using Redis and RQ job queues for background AI workflows.
- Used PostgreSQL for persistent application data and implemented API-driven workflow processing.

Safe Heaven — AI-Powered Crime Detection System

Aug 2023 – Oct 2024

- Developed a computer vision system using Python, OpenCV, Deep Learning, and Flask to analyze live camera feeds for weapons, violence, and suspicious activity.
- Implemented real-time face recognition, automated threat detection, and alerting workflows for continuous monitoring.

Rainfall Forecasting Model

Jul 2025

- Developed a Random Forest Regressor for 7-day rainfall forecasting using historical weather data and Python.
- Engineered 21 time-series lag features and implemented recursive prediction logic for multi-step forecasting.

Experience

AI/ML Intern — iHub School of Learning

- Applied AI/ML concepts to practical projects involving model development, data preprocessing, experimentation, and deployment.
- Worked with Python-based machine learning workflows and contributed to end-to-end AI application development.

Education

Post Graduate Diploma in Artificial Intelligence and Machine Learning

May 2025 – Nov 2025

iHub School of Learning, Edapally, Ernakulam

B.Sc. in Computer Science

Aug 2022 – May 2025

Kerala University, Sree Narayana College, Cherthala

Languages

English, Malayalam